Assignment 3 Output Summary

Repository:
https://github.com/jesseemendozaa/Assignment-3

Purpose:
This report documents representative test cases for countnames.c and shell1.c after the pipe protocol update. The update added a typed message header to the parent-child communication path while preserving the expected visible output.

Build Verification:
```
gcc -Wall -Wextra -g -o countnames countnames.c
gcc -Wall -Wextra -g -o shell1 shell1.c
```

Result:
Both programs compiled successfully with no warnings.

Test Case 1: Direct countnames run on names.txt
Command:
```
./countnames test/names.txt
```

Expected behavior:
- Create one PID.out file with the names and counts found in the file.
- Create one PID.err file with warnings for empty lines.

Observed PID.out contents:
```
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
```

Observed PID.err contents:
```
Warning - file test/names.txt line 2 is empty.
Warning - file test/names.txt line 5 is empty.
```

Notes:
- This confirms direct file parsing still works.
- Empty lines are skipped and reported in the .err file.

Test Case 2: Direct countnames run on testCase2.txt
Command:
```
./countnames test/testCase2.txt
```

Expected behavior:
- Preserve case sensitivity.
- Preserve leading spaces when present in the input.
- Warn about empty lines.

Observed PID.out contents:
```
Jose Mendez: 1
Ana Nguyen: 2
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
```

Observed PID.err contents:
```
Warning - file test/testCase2.txt line 2 is empty.
```

```
Warning - file test/testCase2.txt line 6 is empty.
Warning - file test/testCase2.txt line 10 is empty.
Warning - file test/testCase2.txt line 14 is empty.
```

Notes:
- Ana Nguyen and ANDREW WHITE are counted according to exact text matches.
- The line containing only a space is preserved as " : 1".
- The leading space in " Tim Tran" is preserved.

Test Case 3: shell1 merged run on names.txt and names1.txt
Command entered in shell1:
```
./countnames test/names.txt test/names1.txt
```

Observed merged parent output:
```
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Tom Wu: 3
```

Notes:
- shell1 spawned one child per input file.
- The parent received typed messages from both children and merged duplicate nam
es.
- Tom Wu appears in both files, so the parent combined the counts.

Test Case 4: shell1 merged run on testCase1.txt and testCase3.txt
Command entered in shell1:
```
./countnames test/testCase1.txt test/testCase3.txt
```

Observed merged parent output:
```
ABRAHAM LINCOLN: 1
ABRAHAM RAMIREZ: 1
 ABRAHAM LINCOLN: 1
ADRIAN ADRIAN: 1
ADRIAN: 1
JADA NGUYEn: 1
GEORGE WASHINGTON: 2
JADA NGUYEN: 1
JORGE PEREZ: 1
JESSE MENDOZA: 1
MENDOZA JESSE: 1
Mike Lam: 1
Jimmy Le: 1
J i m m y L e: 1
M ike Lam: 1
JeSsE: 1
JESSE: 1
```

Notes:
- The merged output includes all names from both files.
- Distinct spellings and distinct spacing remain separate entries.
- This confirms that the message header update did not change visible merged res
ults.

Pipe Protocol Verification:
- The code now sends a MessageHeader plus NameCountData together as one fixed-si
ze packet.
- This avoids mismatching a header from one child with payload data from another
 child.
- shell1 checks the message type and payload size before merging counts.

Conclusion:
The updated implementation still compiles and produces the expected direct and m
erged outputs. The message-header enhancement only changes the internal communic
ation format and does not change the intended visible behavior of the assignment
.

Screenshots

Compilation and execution evidence for Assignment 3.

```
JadaNguyen@MacBook group-project-team-12 % cd
JadaNguyen@MacBook ~ % cd "/Users/JadaNguyen/Documents/Assignment-3"
gcc -Wall -Wextra -g -o countnames countnames.c
gcc -Wall -Wextra -g -o shell1 shell1.c

JadaNguyen@MacBook Assignment-3 % ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)
83612.err
83612.out
JadaNguyen@MacBook Assignment-3 % ./countnames test/names.txt
ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)
93839.out
93839.err
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Warning - file test/names.txt line 2 is empty.
Warning - file test/names.txt line 5 is empty.
JadaNguyen@MacBook Assignment-3 % ./countnames test/testCase2.txt
ls -t *.out *.err | head -n 2
cat $(ls -t *.out | head -n 1)
cat $(ls -t *.err | head -n 1)

96416.out
96416.err
Jose Mendez: 1
Ana Nguyen: 2
Charlie Kirk: 1
Barrack Obama: 1
Tommy Obama: 1
 : 1
 Tim Tran: 1
Andrew White: 2
ANDREW WHITE: 1
Nguyen Ana: 1
Warning - file test/testCase2.txt line 2 is empty.
Warning - file test/testCase2.txt line 6 is empty.
Warning - file test/testCase2.txt line 10 is empty.
Warning - file test/testCase2.txt line 14 is empty.
JadaNguyen@MacBook Assignment-3 % ./shell1

% ./countnames test/names.txt test/names1.txt
exit
Nicky: 1
Dave Joe: 2
Yuan Cheng Chang: 3
John Smith: 1
Tom Wu: 3
% %
```